



**School of Computer Science Engineering and Information
Systems**

Winter Semester 2025-2026

Department of Computer Applications

UBCA398J – Project -I / Internship -I

Review -2

Date: 18.3.2026

**Self-Evolving Data Quality Intelligence System for
Machine Learning Pipelines**

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ABSTRACT:

Machine learning models rely heavily on the quality of input data, and poor data quality can significantly degrade predictive performance. This project presents a Self-Evolving Data Quality Intelligence System for Machine Learning Pipelines, designed to identify and mitigate common data quality issues before model training. The proposed system focuses on automated detection and correction of missing and inconsistent values within a healthcare dataset. The Heart Disease dataset was used as the experimental dataset due to the presence of several incomplete attributes and heterogeneous feature types.

The system begins with a data profiling stage that analyzes dataset structure and identifies data quality issues such as missing values and inconsistent data types. A data quality intelligence module then applies context-aware imputation strategies, using median imputation for numerical attributes and mode imputation for categorical attributes. After preprocessing, categorical features are transformed using one-hot encoding to ensure compatibility with machine learning algorithms. A Random Forest classifier is then trained to evaluate the impact of the preprocessing strategy on predictive performance. Model performance is assessed using accuracy and weighted F1-score metrics.

The proposed system includes a feedback-driven mechanism that evaluates model performance and allows the preprocessing strategy to adapt when performance falls below a defined threshold. Experimental results demonstrate that structured data quality improvements lead to measurable performance gains, highlighting the importance of intelligent preprocessing in machine learning workflows.

Keywords: Data Quality, Machine Learning Pipeline, Data Preprocessing, Random Forest, Healthcare Analytics, Missing value imputation, Self-Evolving system

[1.]INTRODUCTION:

- In recent years, machine learning has become an essential tool for extracting meaningful insights from large volumes of data across various domains such as healthcare, finance, and business analytics. The effectiveness of any machine learning model is highly dependent on the quality of the data used during training. Real-world datasets often contain issues such as missing values, inconsistent entries, duplicate records, and mixed data types, which can significantly affect model accuracy and reliability. As a result, ensuring high data quality has become a critical step in the machine learning pipeline.
- Traditional data preprocessing methods are often manual, static, and time-consuming, requiring data scientists to repeatedly clean and transform datasets before training models. These approaches lack adaptability and may not respond effectively when data quality issues evolve over time. To address this challenge, intelligent data quality systems that can automatically detect, correct, and adapt to data inconsistencies are increasingly being explored.
- This project proposes a **Self-Evolving Data Quality Intelligence System for Machine Learning Pipelines** that integrates automated data profiling, preprocessing, and performance feedback mechanisms. The system analyzes incoming datasets, identifies quality issues such as missing values, and applies appropriate correction strategies based on the type of data. A machine learning model is then trained on the processed data to evaluate the effectiveness of the preprocessing steps.

- The system also incorporates a feedback-driven mechanism in which model performance metrics are monitored and used to adjust preprocessing strategies when necessary. By creating a dynamic interaction between data quality management and model performance, the proposed system demonstrates how intelligent preprocessing can enhance the robustness and reliability of machine learning models.
- To validate the adaptability of the proposed system, experiments were conducted on two different datasets: the Heart Disease dataset and the Titanic dataset.

[2.]PROBLEM STATEMENT:

- Machine learning models are increasingly used to support decision-making in critical domains such as healthcare, finance, and business analytics. However, the performance and reliability of these models strongly depend on the quality of the data used for training. In real-world scenarios, datasets often contain issues such as missing values, inconsistent entries, duplicate records, and mixed data types. These data quality problems can lead to biased predictions, reduced model accuracy, and unreliable outcomes.
- Traditional data preprocessing approaches typically involve manual data cleaning and static preprocessing rules. While these methods can address basic data inconsistencies, they often require significant human intervention and may not adapt effectively to varying data conditions. As datasets grow larger and more complex, relying solely on manual preprocessing becomes inefficient and prone to errors.
- The primary challenge addressed in this project is the lack of an intelligent and adaptive system capable of automatically identifying and correcting data quality issues while also evaluating the impact of these corrections on machine learning model performance. Without such a system, organizations may deploy models trained on suboptimal data, leading to inaccurate predictions and reduced trust in machine learning systems.
- Therefore, this project aims to develop a **Self-Evolving Data Quality Intelligence System for Machine Learning Pipelines** that can automatically detect data quality issues, apply appropriate preprocessing strategies, and evaluate the resulting model performance. By incorporating a feedback-driven mechanism, the system seeks to improve the effectiveness of data preprocessing and enhance the overall reliability of machine learning models.

[3.]OBJECTIVES:

The main objective of this project is to design and implement a Self-Evolving Data Quality Intelligence System for Machine Learning Pipelines that can identify and improve data quality before training machine learning models. The project focuses on demonstrating how structured preprocessing and adaptive feedback mechanisms can enhance the performance and reliability of predictive models.

The specific objectives of this project are as follows:

- To analyze real-world datasets and identify common data quality issues such as missing values, inconsistent entries, and mixed data types.

- To develop a data profiling mechanism that evaluates the dataset structure and detects potential quality problems prior to model training.
- To implement automated data preprocessing techniques, including missing value imputation using median for numerical features and mode for categorical features.
- To apply feature transformation techniques such as one-hot encoding to convert categorical data into a format suitable for machine learning algorithms.
- To train and evaluate a machine learning model using a Random Forest classifier to measure the impact of data preprocessing on predictive performance.
- To design a feedback-driven mechanism that monitors model performance metrics and adjusts preprocessing strategies when necessary, demonstrating the self-evolving behavior of the system.
- To compare model performance before and after data quality improvement in order to evaluate the effectiveness of the proposed approach.
- Through these objectives, the project aims to highlight the importance of intelligent data preprocessing and demonstrate how adaptive data quality management can improve the reliability of machine learning pipelines.
- Evaluate the proposed system on multiple datasets to verify its adaptability across different classification problems.

[4.]SCOPE OF THE PROJECT:

- The scope of this project focuses on the development of a **Self-Evolving Data Quality Intelligence System** designed to improve the effectiveness of machine learning pipelines through automated data preprocessing and performance-based feedback. The project primarily addresses common data quality challenges that occur in real-world datasets, such as missing values, inconsistent data formats, and categorical attributes that require transformation before model training.
- The system is implemented using the Python programming language along with widely used data science libraries including Pandas, NumPy, and Scikit-learn. The project uses the Heart Disease dataset and the Titanic dataset to demonstrate its applicability across multiple classification tasks.
- The scope of the project includes data profiling to analyze dataset structure, detection of missing values, implementation of appropriate imputation techniques, and transformation of categorical variables into numerical formats using encoding methods. A Random Forest classifier is used as the predictive model to evaluate the impact of preprocessing on model performance.
- Additionally, the project introduces a feedback mechanism that evaluates model performance metrics and allows the system to adjust preprocessing strategies when required, representing the self-evolving capability of the system. The experimental

evaluation compares model performance before and after data quality improvement to highlight the effectiveness of the proposed approach.

- However, the project is limited to structured tabular datasets and focuses primarily on handling missing data and basic preprocessing tasks. Advanced data quality issues such as large-scale streaming data, real-time pipeline integration, and deep learning-based data correction are beyond the scope of this implementation.

[5.]PROPOSED SYSTEM:

- The proposed system presents a **Self-Evolving Data Quality Intelligence System for Machine Learning Pipelines** designed to improve the reliability and performance of machine learning models through intelligent data preprocessing and adaptive model optimization. The system integrates data profiling, automated data cleaning, feature transformation, machine learning model training, and hyperparameter tuning within a unified workflow.
- The system begins with a **data ingestion module**, where datasets are loaded and prepared for analysis. In this project, two datasets were used to validate the adaptability of the proposed system: the **Heart Disease dataset** and the **Titanic dataset**. Using multiple datasets helps demonstrate that the system can handle different classification problems across domains.
- Next, a **data profiling stage** analyzes the dataset structure to detect common data quality issues such as missing values, duplicate records, and inconsistent data types. Identifying these issues allows the system to apply appropriate preprocessing strategies.
- The **data quality intelligence module** then performs automated preprocessing. Missing values in numerical attributes are handled using **median imputation**, while categorical attributes are handled using **mode imputation**. These techniques ensure that missing values are replaced using statistically meaningful approaches. Additionally, categorical features are transformed into numerical format using **one-hot encoding**, enabling compatibility with machine learning algorithms.
- After preprocessing, the system trains a **Random Forest classifier**, which was selected due to its robustness and ability to handle complex relationships within structured data. To further improve model performance, **hyperparameter tuning is performed using GridSearchCV**, which searches for the optimal configuration of model parameters such as the number of trees, tree depth, and minimum samples required for node splitting.
- The final stage involves **model evaluation**, where performance is measured using multiple metrics including **Accuracy, Precision, Recall, and F1-score**. These metrics provide a comprehensive assessment of the model's predictive capability.
- By combining automated data quality management, machine learning, and adaptive parameter optimization, the proposed system demonstrates how intelligent preprocessing and model optimization can enhance the reliability and effectiveness of machine learning pipelines across multiple datasets

[6.]LITERATURE SURVEY: (minimum 15 papers)

S.NO	TITLE	MERITS	DEMERITS
1	Data Cleaning: Problems and Current Approaches (Rahm & Do, 2000)	Provides comprehensive overview of data cleaning techniques and challenges	Mainly theoretical, lacks automated adaptive solutions
2	Data Mining: Concepts and Techniques (Han, Kamber & Pei, 2011)	Strong foundation for preprocessing, feature engineering and ML techniques	Focuses broadly on data mining rather than automated data quality systems
3	A Survey of Data Quality Management Techniques	Explains methods for detecting and improving data quality	Does not integrate machine learning feedback mechanisms
4	Handling Missing Data in Machine Learning	Discusses various imputation techniques such as mean, median and mode	Performance varies depending on dataset characteristics
5	Random Forests (Breiman, 2001)	Introduces a powerful ensemble learning algorithm with high accuracy	Computational cost increases with large datasets
6	Data Preprocessing in Data Mining	Demonstrates importance of preprocessing for improving model performance	Limited discussion on dynamic preprocessing strategies
7	Automated Machine Learning: Methods, Systems and Challenges	Introduces automated ML pipelines that reduce manual effort	Focuses more on model selection than data quality improvement
8	Feature Engineering for Machine Learning (Alice Zheng, 2018)	Shows how feature transformation improves model performance	Requires domain knowledge and manual effort
9	Data Quality Assessment for Big Data Applications	Explores techniques for managing data quality in large datasets	Complex implementation for smaller systems
10	Machine Learning Pipelines for Predictive Modeling	Explains structured ML workflow including preprocessing and evaluation	Limited focus on intelligent adaptive data cleaning
11	Handling Imbalanced Data in Classification Problems	Provides strategies to improve classification performance in skewed datasets	Requires additional tuning and parameter optimization
12	Healthcare Data Mining and Predictive Analytics	Demonstrates applications of ML in healthcare diagnosis	Strongly dependent on high quality medical data
13	Missing Data Imputation in Healthcare Datasets	Presents methods for handling missing medical records	Does not include adaptive feedback learning mechanisms

14	Intelligent Data Processing Systems	Introduces concept of automated data processing systems	Implementation complexity can be high
15	Adaptive Data Processing for Intelligent Systems	Demonstrates feedback driven systems for improving data processing pipelines	Requires continuous monitoring and system tuning

[6.1] FINDINGS IN LITERATURE SURVEY:

- The literature survey highlights the significant role of data quality and preprocessing in improving the performance and reliability of machine learning models. Several studies emphasize that real-world datasets often contain issues such as missing values, inconsistent records, and noise, which can negatively impact model accuracy and predictive capability. Effective data preprocessing techniques, including data cleaning, feature transformation, and missing value imputation, are therefore considered essential steps in the machine learning pipeline.
- Existing research also demonstrates that machine learning algorithms such as Random Forest classifiers are widely used due to their robustness, high accuracy, and ability to handle complex relationships within data. However, many traditional data preprocessing methods rely on static rules and manual intervention, which limits their adaptability when dealing with evolving datasets.
- Another key observation from the literature is that automated machine learning systems focus primarily on model selection and optimization while often overlooking intelligent data quality management. This creates a gap where preprocessing steps remain largely manual and disconnected from model performance evaluation.
- Furthermore, several studies indicate that healthcare datasets are particularly sensitive to data quality issues because incomplete or inconsistent medical records can lead to unreliable diagnostic predictions.
- Based on these findings, it becomes clear that integrating automated data quality management with machine learning pipelines can significantly enhance predictive performance. This insight forms the foundation of the proposed self-evolving data quality intelligence system, which combines data profiling, adaptive preprocessing, and performance-based feedback mechanisms to improve overall model reliability.

[7]. METHODOLOGY:

- The methodology of the proposed system focuses on integrating data quality

management with machine learning techniques to improve predictive performance. The system follows a structured workflow consisting of data ingestion, data profiling, preprocessing, feature transformation, model training, hyperparameter tuning, and model evaluation. This approach ensures that data quality issues are addressed before training machine learning models.

- The first stage is data ingestion, where the datasets are loaded into the system using Python-based data analysis libraries such as Pandas. In this project, two datasets were used to evaluate the system: the Heart Disease dataset and the Titanic dataset. Using multiple datasets helps demonstrate the adaptability of the proposed system across different classification problems.
- The next stage is data profiling, which analyzes the dataset structure and identifies potential data quality issues. During this stage, the system examines the dataset for missing values, duplicate records, and inconsistent data types. Understanding these issues is important for selecting appropriate preprocessing techniques.
- Following data profiling, the system performs data preprocessing and data cleaning. Missing values are handled using statistically appropriate methods. Numerical attributes are imputed using the median value, while categorical attributes are imputed using the mode value. This approach helps maintain the natural distribution of the data and avoids introducing unrealistic values.
- After handling missing values, the system performs feature transformation. Categorical attributes are converted into numerical format using one-hot encoding so that they can be used effectively by machine learning algorithms.
- The cleaned and transformed dataset is then divided into training and testing sets using a train-test split method. A Random Forest classifier is used as the predictive model because of its robustness, ability to handle non-linear relationships, and effectiveness with structured tabular data.
- To further improve model performance, the system performs hyperparameter tuning using GridSearchCV. This step evaluates multiple combinations of model parameters such as the number of trees, maximum tree depth, and minimum samples required for node splitting. The optimal parameter configuration is selected based on cross-validation performance.
- Finally, the system performs model evaluation. The trained model is evaluated using multiple metrics including Accuracy, Precision, Recall, and F1-score. These metrics

provide a comprehensive understanding of model performance and classification effectiveness.

- Through this methodology, the proposed system demonstrates how structured data preprocessing, machine learning, and parameter optimization can be combined to build a reliable and adaptive machine learning pipeline.

[8.] SOFTWARE REQUIREMENTS:

The software requirements define the functional and non-functional capabilities necessary for the successful implementation of the proposed system. These requirements describe how the system should operate and the conditions under which it must perform efficiently.

8.1 Functional Requirements

Functional requirements describe the specific operations and tasks that the system must perform.

1. The system should allow users to load and analyze datasets such as the Heart Disease dataset and the Titanic dataset.
2. The system should perform data profiling to analyze dataset characteristics, including detection of missing values, duplicate records, and data types.
3. The system should automatically identify and handle missing values using appropriate imputation techniques such as median imputation for numerical attributes and mode imputation for categorical attributes.
4. The system should transform categorical variables into numerical format using encoding techniques such as one-hot encoding.
5. The system should train a machine learning model using the Random Forest classifier to perform classification tasks.
6. The system should perform hyperparameter tuning using GridSearchCV to determine the optimal configuration of the machine learning model.

7. The system should evaluate model performance using metrics such as Accuracy, Precision, Recall, and F1-score.
 8. The system should display results including model performance metrics, confusion matrix, and comparison graphs.
 9. The system should allow comparison of model performance across different preprocessing stages such as before cleaning, after cleaning, and after hyperparameter tuning.
-

8.2 Non-Functional Requirements

Non-functional requirements describe the quality attributes and constraints of the system.

1. Performance:
The system should process datasets efficiently and perform model training and evaluation within a reasonable time frame.
2. Reliability:
The system should consistently provide accurate preprocessing and model evaluation results without system failures.
3. Usability:
The system should be easy to use and allow users to execute the machine learning pipeline through a simple and clear workflow.
4. Scalability:
The system should be capable of handling datasets of moderate size and be adaptable to other structured datasets.
5. Maintainability:
The system should be designed in a modular way so that components such as preprocessing techniques or machine learning models can be easily modified or extended.
6. Portability:

The system should be executable on common computing environments that support Python and machine learning libraries.

7. Security:

The system should ensure that dataset inputs are handled securely without causing unauthorized modifications to system resources.

[9.] SYSTEM ARCHITECTURE:

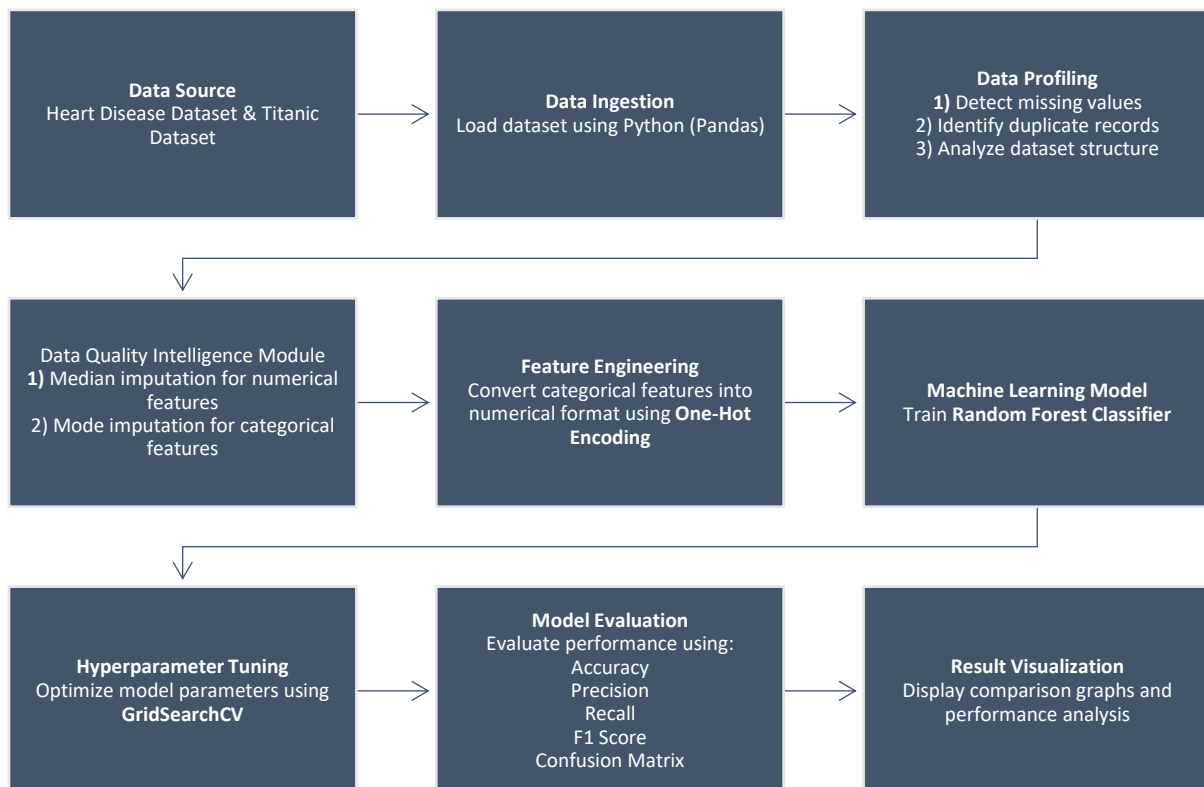


Figure 9.1: System Architecture of the Proposed Self-Evolving Data Quality Intelligence System

[10]. UML DIAGRAMS:

Use Case:

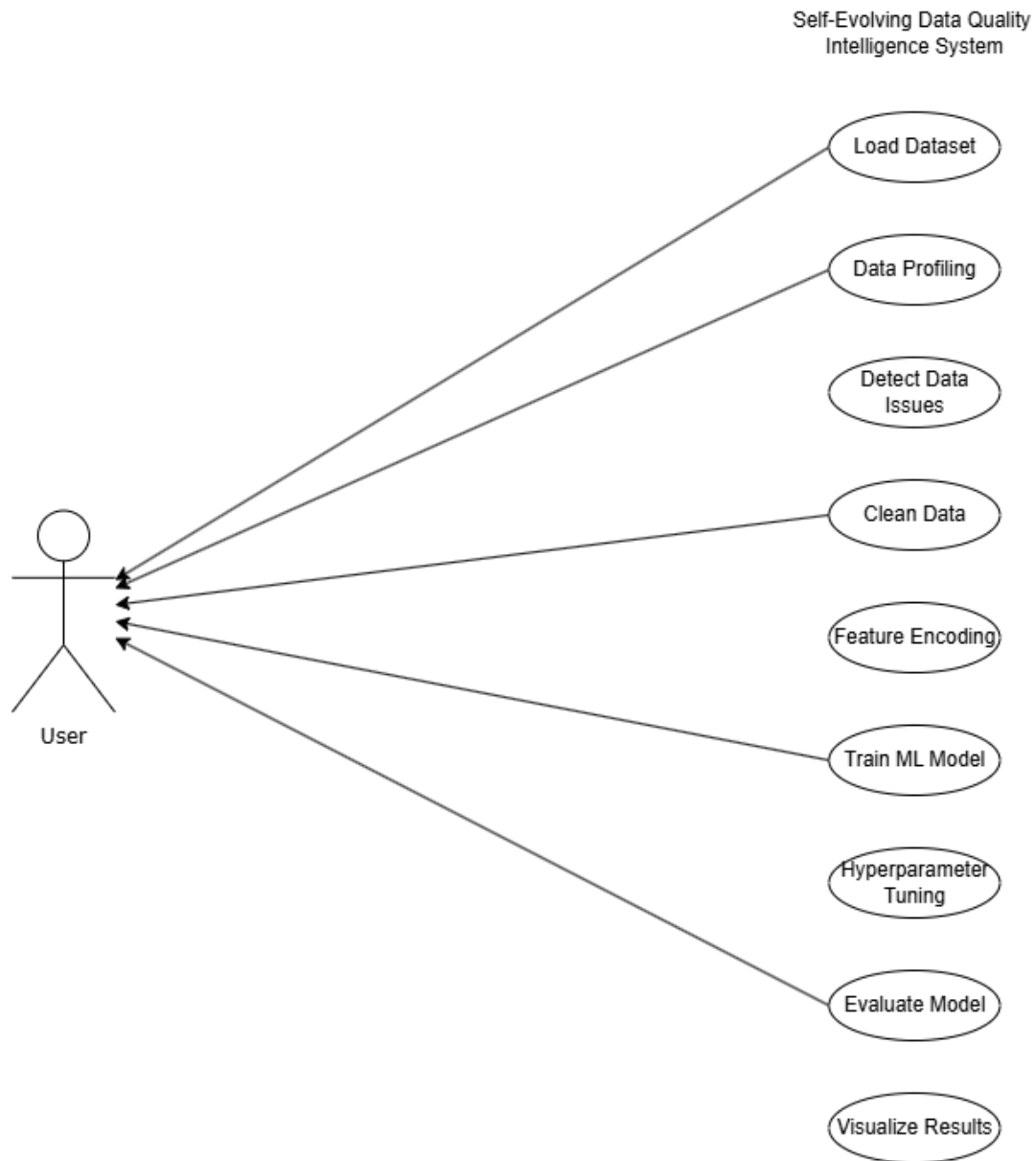


Figure 10.1: Use Case Diagram of the Proposed Self-Evolving Data Quality Intelligence System

Class Diagrams :

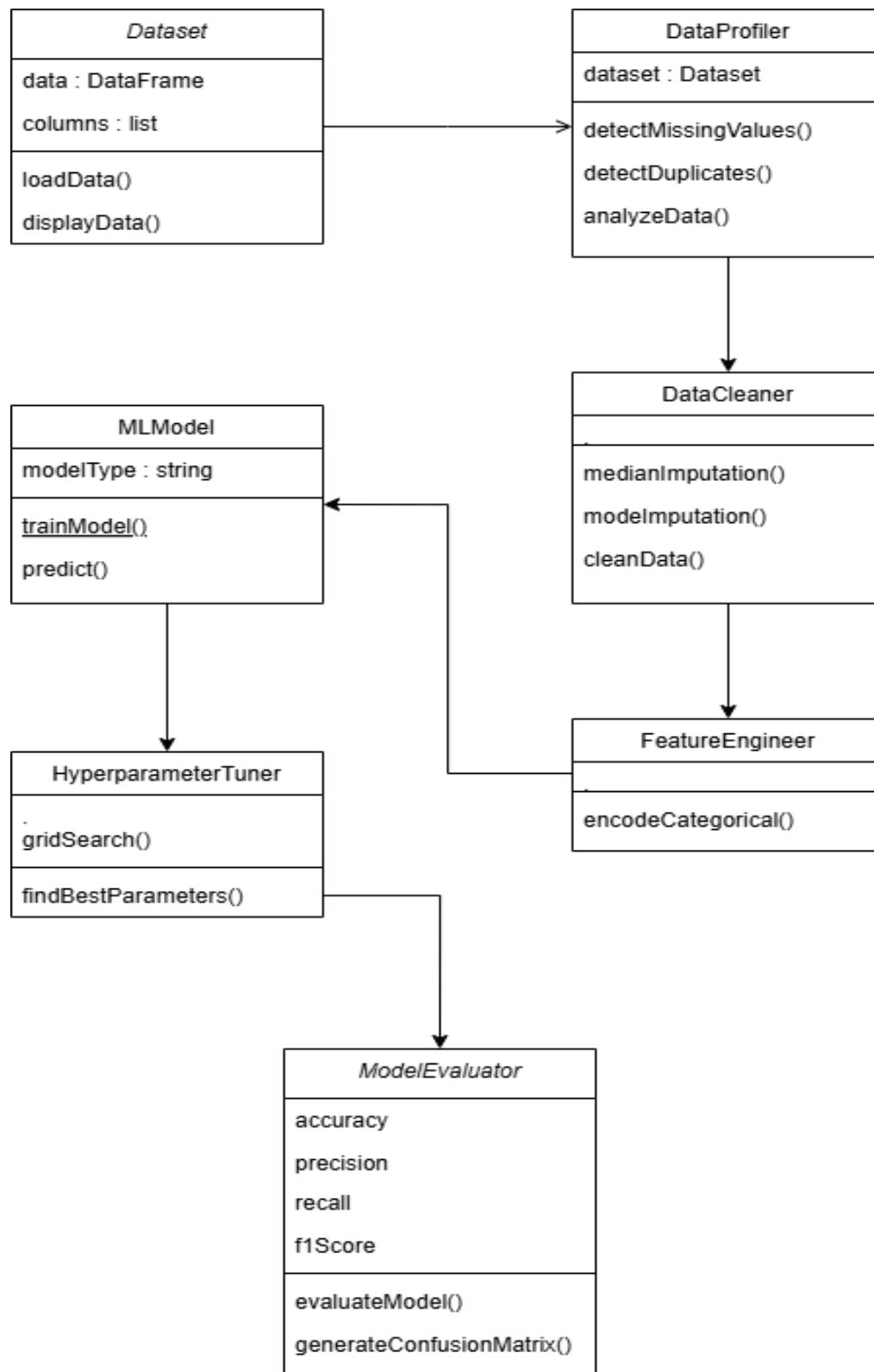


Figure 10.2: Class Diagram representing the structural components of the machine learning pipeline

[11]. IMPLEMENTATION:

- The proposed system was implemented using Python with libraries such as Pandas, NumPy, and Scikit-learn. The development was carried out in two stages: experimentation using Jupyter Notebook and deployment using Streamlit.
- Initially, the datasets were analyzed and preprocessed in Jupyter Notebook. Data cleaning techniques such as handling missing values using median and mode imputation were applied. Categorical features were encoded using one-hot encoding.
- A Random Forest classifier was used for model training and evaluation. Performance metrics including accuracy, precision, recall, and F1-score were calculated. Hyperparameter tuning was performed using GridSearchCV to optimize model performance.
- The finalized pipeline was then integrated into a Streamlit application, allowing users to upload datasets and visualize results interactively. The system demonstrates data quality improvement and model performance comparison across different stages.

11.1. PROJECT PLAN:

- ☐ Problem Identification
 - Identify issues in data quality affecting ML performance
- ☐ Dataset Collection
 - Heart Disease dataset
 - Titanic dataset
- ☐ Data Preprocessing
 - Handle missing values
 - Remove irrelevant features
 - Encode categorical variables
- ☐ Model Development
 - Train Random Forest classifier
- ☐ Model Evaluation
 - Measure accuracy, precision, recall, F1-score
- ☐ Hyperparameter Tuning
 - Optimize model using GridSearchCV
- ☐ System Development
 - Build interactive interface using Streamlit
- ☐ Testing and Validation
 - Compare performance before and after preprocessing
- ☐ Result Analysis
 - Interpret results and system behavior

11.2 SAMPLE CODE:

Code 1: Data Cleaning (Jupyter):

```
df_cleaned = df.copy()

for col in df_cleaned.columns:
    if col != target:
        if df_cleaned[col].dtype == 'object':
            df_cleaned[col] = df_cleaned[col].fillna(df_cleaned[col].mode()[0])
        else:
            df_cleaned[col] = df_cleaned[col].fillna(df_cleaned[col].median())
```

Code 2: Encoding:

```
df_encoded = pd.get_dummies(df_cleaned, drop_first=True)
```

Code 3: Model Training:

```
model = RandomForestClassifier(random_state=42)
model.fit(X_train, y_train)
```

Code 4: Evaluation:

```
accuracy = accuracy_score(y_test, pred)
f1 = f1_score(y_test, pred, average='weighted')
```

Code 5: Hyperparameter Tuning:

```
grid = GridSearchCV(RandomForestClassifier(), param_grid, cv=3)
grid.fit(X_train, y_train)
```

Code 6: Streamlit Upload:

```
file = st.file_uploader("Upload CSV Dataset")
```

11.3 SAMPLE SCREEN SHOTS

Jupyter Screeshots:

Dataset preview:

Heart Disease:

	id	age	sex	dataset	cp	trestbps	chol	fbs	restecg	thalch	exang	oldpeak	slope	ca	thal	num
0	1	63	Male	Cleveland	typical angina	145.0	233.0	True	lv hypertrophy	150.0	False	2.3	downsloping	0.0	fixed defect	0
1	2	67	Male	Cleveland	asymptomatic	160.0	286.0	False	lv hypertrophy	108.0	True	1.5	flat	3.0	normal	2
2	3	67	Male	Cleveland	asymptomatic	120.0	229.0	False	lv hypertrophy	129.0	True	2.6	flat	2.0	reversable defect	1
3	4	37	Male	Cleveland	non-anginal	130.0	250.0	False	normal	187.0	False	3.5	downsloping	0.0	normal	0
4	5	41	Female	Cleveland	atypical angina	130.0	204.0	False	lv hypertrophy	172.0	False	1.4	upsloping	0.0	normal	0

Titanic Survivor:

[3]:	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Missing values:

Titanic:

```
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age             177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin           687
Embarked         2
dtype: int64
```


Model result:

Before Cleaning: Titanic:

```
Before Cleaning
Accuracy: 0.659217877094972
Precision: 1.0
Recall: 0.17567567567567569
F1 Score: 0.2988505747126437
```

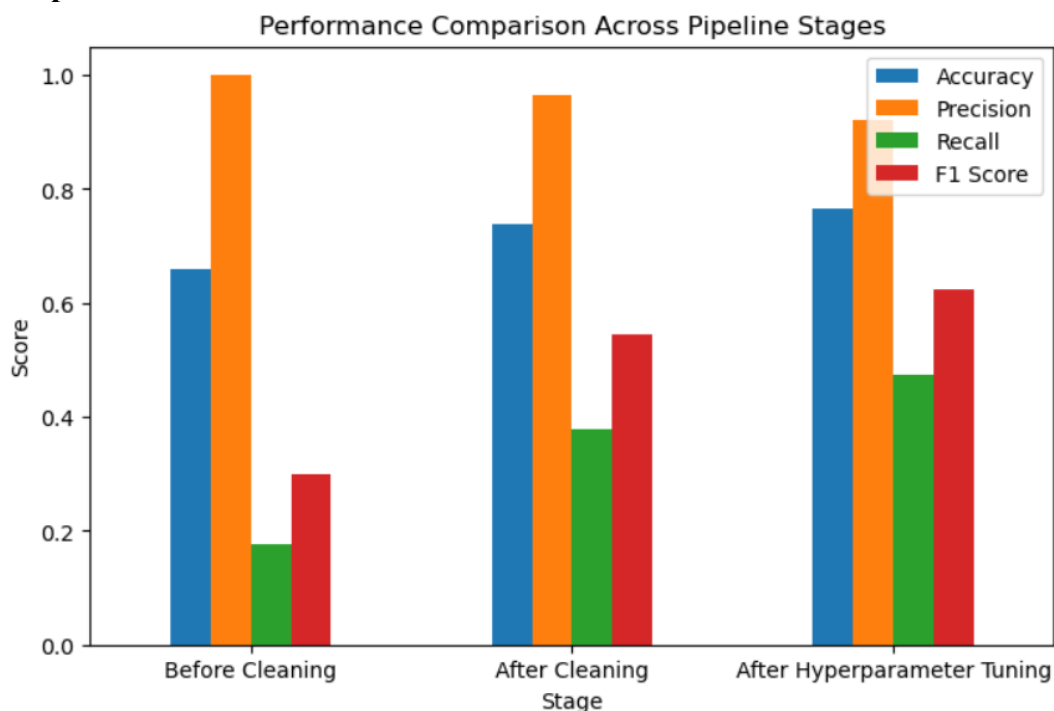
After cleaning:

```
After Cleaning
Accuracy: 0.7374301675977654
Precision: 0.9655172413793104
Recall: 0.3783783783783784
F1 Score: 0.5436893203883495
```

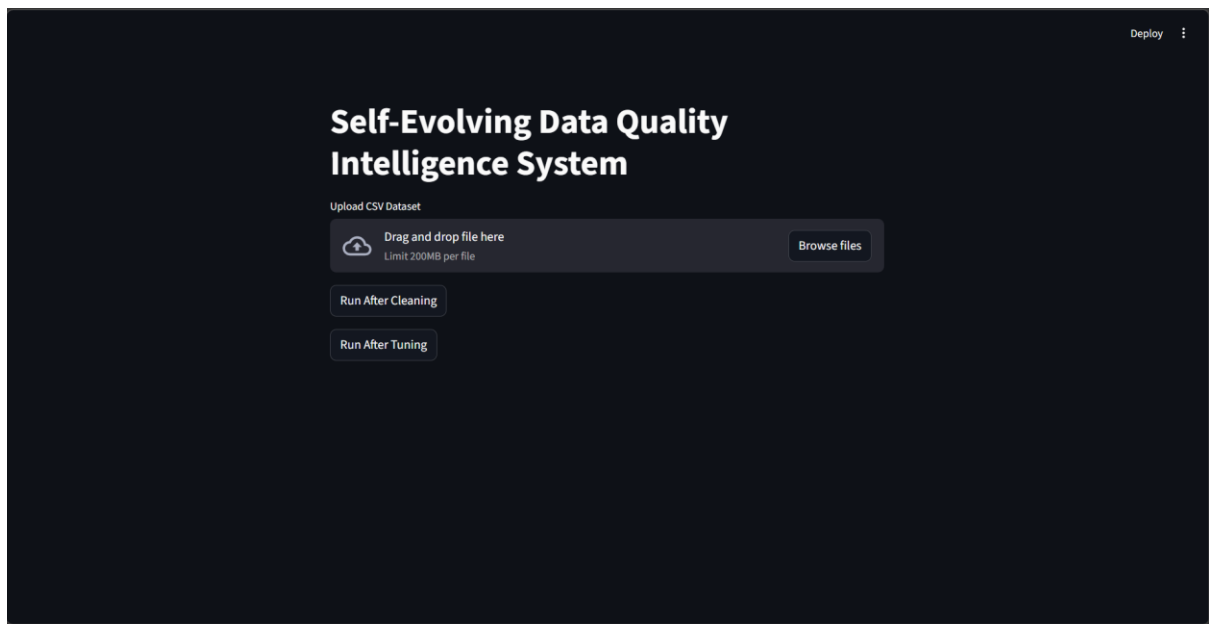
Hyperparameter tuning output:

```
Best Parameters: {'max_depth': 15, 'min_samples_split': 5, 'n_estimators': 100}
After Hyperparameter Tuning
Accuracy: 0.7653631284916201
Precision: 0.9210526315789473
Recall: 0.47297297297297297
F1 Score: 0.625
```

Graphs:



STREAMLIT SCREENSHOTS:



Dataset Preview (Before Cleaning)

	age	sex	cp	trestl	chol	fbs	restecg	thalch	exang	oldpeak
0	63	Male	typical angina	145	233	☒	lv hypertrophy	150	☐	2.3
1	67	Male	asymptomatic	160	286	☐	lv hypertrophy	108	☒	1.5
2	67	Male	asymptomatic	120	229	☐	lv hypertrophy	129	☒	2.6
3	37	Male	non-anginal	130	250	☐	normal	187	☐	3.5
4	41	Female	atypical angina	130	204	☐	lv hypertrophy	172	☐	1.4

Missing Values Before Cleaning

	0
age	0
sex	0
cp	0
trestbps	59
chol	30
fbs	90
restecg	2
thalch	55
exang	55
oldpeak	62

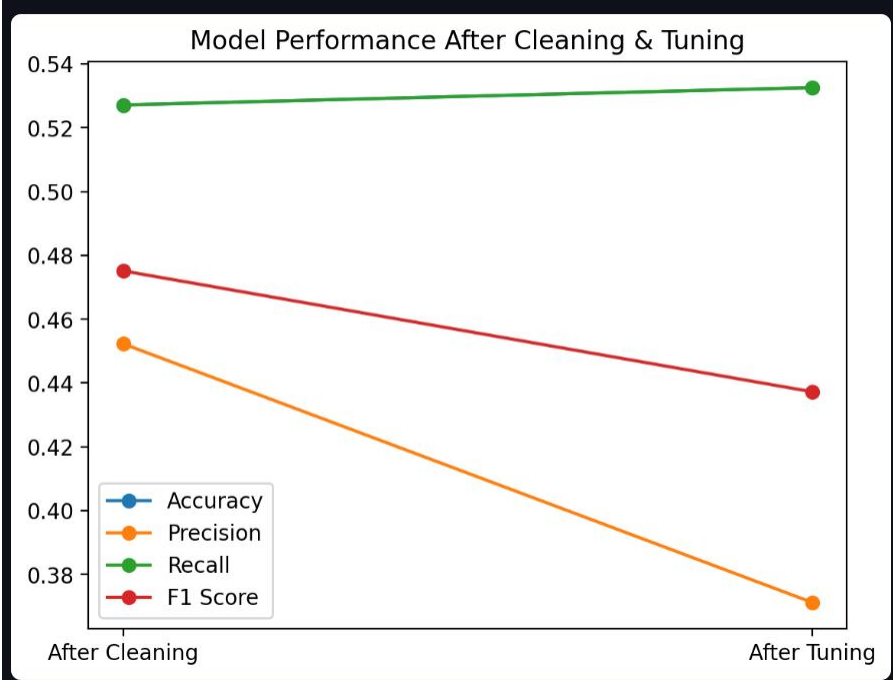
Missing Values After Cleaning

	0
age	0
sex	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalch	0
exang	0
oldpeak	0

Model Performance

	Stage	Accuracy	Precision	Recall	F1 Score
0	After Cleaning	0.5272	0.4523	0.5272	0.4752
1	After Tuning	0.5326	0.3712	0.5326	0.4372

Performance Graph



[12]SUMMARY:

This project presents the development of a **Self-Evolving Data Quality Intelligence System** aimed at improving the effectiveness of machine learning pipelines through automated data preprocessing and performance evaluation. The system addresses common data quality issues such as missing values, inconsistent data, and irrelevant features, which can negatively impact model performance.

The implementation was carried out using Python, with experimentation performed in Jupyter Notebook and deployment through a Streamlit-based interactive interface. Two datasets, namely the Heart Disease dataset and the Titanic dataset, were used to validate the system's effectiveness across different domains.

Data preprocessing techniques including missing value imputation, feature encoding, and removal of unnecessary attributes were applied to enhance dataset quality. A Random Forest classifier was used for model training, and performance was evaluated using accuracy, precision, recall, and F1-score. Hyperparameter tuning was also incorporated to optimize model performance.

The results demonstrate that while accuracy improvements may vary, the system consistently enhances data quality and ensures stable and reliable model performance. The integration of preprocessing and evaluation within a single pipeline highlights the importance of data quality in machine learning systems.

Overall, the project successfully showcases an intelligent and adaptive approach to data preprocessing, contributing to more robust and trustworthy machine learning models.

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